Scale Drawings

Teacher's Edition

The Period

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must Find the scale factor between two figures (k = copy ÷ original, in either direction of size) and use a scale factor or a scale to find a length in either direction; recognise that angles are unchanged. (MA.7.GR.1.5, clarifications 1 and 3.)

Must not Perimeter and area under scaling are 5.02–03 (k^2); floor plans are 5.04; dilations on the coordinate plane are MA.8.GR.2.2; similar triangles by name are Unit 6.

Target I can find the scale factor between two figures, and use a scale to find a real length from a drawing or a drawing length from the real one.

MTR **MTR.2.1** — Notes I: the same pair of figures as a picture, two side lengths and one number k · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.5.1** — boards 1–3: one question three times

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	13 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A scale factor is the one number every length is multiplied by. Copy divided by original.
2. Less than 1 shrinks. More than 1 enlarges. The angles never change.
3. To the copy, multiply. Back to the original, divide.

Slide by Slide

1. **Scale Drawings.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the lesson in miniature: a scale is a ratio, and the first move is always to read it per 1. *1 retrieves 4.06 — multiply before adding; the test was yesterday, so the first band reaches to the last lesson (yesterday). 2 retrieves 4.05 — a perfect square's root; 5.02–03 uses it to find k from two areas (last week). 3 retrieves 3.08 — reading scientific notation back (last unit). 4 retrieves MA.6.AR.3.2 — a ratio as a unit rate: 12 ÷ 3. Today's scale is exactly this, read per 1 (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. −5 2. 9 3. 2,500 4. 4 km
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the figure and the line.** Two rectangles, 6 by 4 and 9 by 6.
6. **Notes.** The direction is fixed all unit: copy ÷ original.
7. **Notes.** From the original to the copy, multiply; from the copy back to the original, divide.
8. **Example 1 — Find k from two figures.** Ask which side matches which before anyone divides: 15 matches 6, 10 matches 4.
9. **Example 1 — Worked.** 15 ÷ 6 = 2.5 and 10 ÷ 4 = 2.5: the same k from both pairs, so it is a copy and k = 2.5.
10. **Your Turn — Same steps, your numbers. Boards up when done.** k = 1.5 from both pairs (9 ÷ 6 and 12 ÷ 8).
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Use a scale on a map.** Ask what the scale says per 1 inch before anyone multiplies.
13. **Example 2 — Worked.** Every inch is 20 miles, so 3.5 inches is 3.5 × 20 = 70 miles.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 63 km: 4.2 × 15.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** A 2 cm side becomes 6 cm in the copy. What is the scale factor? → ***k = 3*** *⅓ — divided original by copy [B1G-M 7.GR.1.5: the proportion set up backwards]; 4 — subtracted [the additive error]*
18. **18–19. Board 2.** An 8 cm side becomes 4 cm in the copy. What is the scale factor? → ***k = ½*** *2 — divided original by copy; 'k is always bigger than 1' [B1G-M 7.GR.1.5 named misconception]*
20. **20–21. Board 3.** A 12 cm side becomes 9 cm in the copy. What is the scale factor? → ***k = ¾*** *1⅓ — original ÷ copy [7.GR.1.5]; 3 — subtracted [the additive error]*
22. **22–23. Board 4.** A side of 5 cm becomes 2 cm in the copy. What is the scale factor? → ***B — 0.4*** *(A) original ÷ copy: 5 ÷ 2 [B1G-M 7.GR.1.5: the proportion set up backwards; 'k is always greater than 1']; (C) 'the same shape, so k = 1' [Math Nation 5.3: a scale factor of 1 is the same size, not the same shape]; (D) 5 − 2: the additive error — 'to get from 5 to 2, take away 3' [B1G-M 7.AR.3.3: 'just add' as a conversion] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. The scale factor is 3.5. A side of the original is 6 cm. How long is that side in the copy? → ***21 cm*** *1.7 — divided by k going to the copy [Notes III: to the copy, multiply]*

26–27. Board 6. A drawing has a 40° angle. The real figure is a scale copy at k = 2.5. What is the angle in the real figure? → ***A — 40°*** *(B) 40 + 2.5: the additive error [B1G-M]; (C) 40 ÷ 2.5: the angle divided [7.GR.1.5]; (D) 40 × 2.5: the angle scaled like a length [7.GR.1.5: a scale factor multiplies lengths; the shape — its angles — is what stays the same] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A triangle has sides 6, 8 and 10. Another has sides 9, 12 and 15; a third has sides 9, 12 and 14. Which one is a scale copy of the first? → ***The 9-12-15 triangle (k = 1.5); the 9-12-14 one is not a copy*** *'both' — checked one pair and stopped [Notes I: the same k from every side]; 'neither' — compared 9 to 10 instead of corresponding sides*

30–31. Board 8. Dev, who is 13, draws a 12-ft wall as 3 in long. How long is a 20-ft wall on his drawing? → ***5 in*** *The 13 is not needed. 80 — multiplied by 4 going to the drawing [Notes III: back to the drawing, divide]; 11 — 20 − 12 + 3, the additive error*

32–33. Board 9. Priya says the scale factor from a 6 cm side to its 3 cm copy is 2. Is she right? Explain, and give the scale factor. → ***No. k = 3 ÷ 6 = ½: the copy is smaller, so k is under 1.*** *'Yes, because 6 is 2 times 3' — the proportion backwards [B1G-M 7.GR.1.5]; 'No, it is 3' — subtracted*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. ⅓ — divided original by copy [B1G-M 7.GR.1.5: the proportion set up backwards]

Board 2. 2 — divided original by copy

Board 3. 1⅓ — original ÷ copy [7.GR.1.5]

Board 4. original ÷ copy: 5 ÷ 2 [B1G-M 7.GR.1.5: the proportion set up backwards

Board 5. 1.7 — divided by k going to the copy [Notes III: to the copy, multiply]

Board 6. 40 + 2.5: the additive error [B1G-M]

Board 7. 'both' — checked one pair and stopped [Notes I: the same k from every side]

Board 8. 80 — multiplied by 4 going to the drawing [Notes III: back to the drawing, divide]

Board 9. 'Yes, because 6 is 2 times 3' — the proportion backwards [B1G-M 7.GR.1.5]

Materials Whiteboards and markers. Two squares of different sizes cut from paper, for board 6's split: four right angles each. Calculators are allowed on everything and printed on nothing.

Scale Factor: Perimeter and Area

Teacher's Edition

The Period

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must Find the perimeter and area of a scale copy from the original and k (perimeter × k, area × k^2), and find k from two areas when the ratio is a perfect square. (MA.7.GR.1.5, clarification 2.)

Must not Area ratios that are not perfect squares (an approximate root); the area of a figure the class has no formula for; area and perimeter in different units (5.04 and 5.08–09).

Target I can find the perimeter and the area of a scale copy from the original and k, and find k from two areas.

MTR **MTR.2.1** — Notes II: the same copy as a grid picture, as 15 → 60, and as $k^2 = 4$ · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote on k against k^2 · **MTR.5.1** — boards 1–2: the same k on the perimeter, then the area, and the room is asked why the second answer is bigger

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The perimeter is lengths, so it is multiplied by k.
2. Area is two lengths multiplied, so it is multiplied by k twice: k squared.
3. From two areas, k is the square root of the ratio.

Slide by Slide

1. **Scale Factor: Perimeter and Area.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the figure Notes I and II scale: 3 by 5, P = 16, A = 15. *1 retrieves 5.01 — copy ÷ original (yesterday). 2 retrieves 4.05 — a perfect square's root; Notes III finds k this way (last week). 3 retrieves 3.01 — powers; k^2 is a power (last unit). 4 retrieves MA.6.GR.2.1 — perimeter is the sum of the sides, area is length × width; today both are scaled (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. k = 2.5 2. 12 3. 9 and 27 4. P = 16 cm, A = 15 cm^2
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the figure and the two lines.** The 3 by 5 from the warm-up at k = 2: 6 by 10.
6. **Notes.** Area is length × width, and both were multiplied by k, so the area is multiplied by k twice: 15 → 60, four times as much, not twice.
7. **Notes.** Run it backwards.
8. **Example 1 — Perimeter and area from k.** Ask which rule each part uses before anyone multiplies: perimeter × k, area × k^2.
9. **Example 1 — Worked.** P = 12 → 36 (× 3); A = 8 → 72 (× 9).
10. **Your Turn — Same steps, your numbers. Boards up when done.** P 16 → 64, A 15 → 240.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — k from two areas, then a side.** Ask what 180 ÷ 20 is, and whether that number is k.
13. **Example 2 — Worked.** 180 ÷ 20 = 9 is k^2, so k = 3, and the side is 12.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 300 ÷ 12 = 25, k = 5, the side is 10.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. A figure with perimeter 14 cm is scaled by k = 2. What is the copy's perimeter? → ***28 cm*** *56 — perimeter × k^2 [Notes I: a perimeter is lengths, × k]; 16 — added k [the additive error]*

18–19. Board 2. A figure with area 12 cm^2 is scaled by k = 2. What is the copy's area? → ***48 cm^2*** *24 — area × k [B1G-M 7.GR.1.5 named misconception: scaling area like length]*

20–21. Board 3. A figure with area 48 cm^2 is scaled by k = ½. What is the copy's area? → ***12 cm^2*** *24 — area × k [7.GR.1.5: area × k^2]; 96 — divided by ½ as if k were 2 [5.01: k under 1 shrinks]*

22–23. Board 4. A figure with area 10 cm^2 is scaled by k = 3. What is the copy's area? → ***A — 90 cm^2*** *(B) area × 2k: 'two lengths, so twice k' [Notes II: k × k, not k + k]; (C) area × k: the guide's named misconception [B1G-M 7.GR.1.5: scaling area like length]; (D) 10 + 3: the additive error [B1G-M] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. A figure with area 8 cm² is scaled by k = 1.5. What is the copy's area? → ***18 cm²*** *12 — area × k [7.GR.1.5]; 24 — area × 2k [Notes II: k × k]*

26–27. Board 6. A figure's perimeter is 20 cm; its copy's perimeter is 60 cm. What is the area ratio, copy to original? → ***D — 9*** *(A) 2k: 'area has two dimensions, so double it' [Notes II]; (B) the perimeter ratio handed in as the area ratio [Notes I and II: perimeter gives k, area is k²]; (C) 60 − 20: the additive error [B1G-M] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A triangle with sides 6, 8 and 10 cm is scaled by k = ½. Its area is 24 cm². Find the copy's perimeter and area. → ***P = 12 cm, A = 6 cm²*** *P 12, A 12 — the area halved like a length [7.GR.1.5]; P 6, A 6 — both quartered*

30–31. Board 8. A 4 in by 6 in photo, taken 3 years ago, is enlarged at k = 2.5. What is the enlargement's area? → ***150 in²*** *The 3 years is not needed. 60 — area × k [7.GR.1.5]; 120 — 24 × 5, k doubled [Notes II]*

32–33. Board 9. Jo says that doubling every side of a rectangle doubles its area. Explain why not, with a counterexample. → ***No: 2 by 7 doubled is 4 by 14, area 14 → 56 — × 4, as both lengths doubled.*** *'Yes, because every side doubles' — scaling area like length [B1G-M 7.GR.1.5]; 'No, it is k²' with no numbers — the rule without the reason*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 56 — perimeter × k² [Notes I: a perimeter is lengths, × k]

Board 2. 24 — area × k [B1G-M 7.GR.1.5 named misconception: scaling area like length]

Board 3. 24 — area × k [7.GR.1.5: area × k²]

Board 4. area × 2k: 'two lengths, so twice k' [Notes II: k × k, not k + k]

Board 5. 12 — area × k [7.GR.1.5]

Board 6. 2k: 'area has two dimensions, so double it' [Notes II]

Board 7. P 12, A 12 — the area halved like a length [7.GR.1.5]

Board 8. 60 — area × k [7.GR.1.5]

Board 9. 'Yes, because every side doubles' — scaling area like length [B1G-M 7.GR.1.5]

Materials Whiteboards and markers. A sheet of grid paper under the document camera for Notes II's four-copies picture, drawn live if the slide's grid is not enough. Calculators are allowed on everything and printed on nothing.

Real-World Scale Drawings

Teacher's Edition

The Period

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must Read a scale such as ¼ in : 1 ft or 1 : 50 as a rate, find real lengths and drawing lengths from it in both directions, and find a real area from the real dimensions. (MA.7.GR.1.5, clarifications 1 and 3; B1G-M's wall example.)

Must not Converting the real area into another system (5.08–09); drawing a plan to scale with a ruler (not a FAST shape — the audit's ruler items are out); a scale factor between two drawings of one object (5.01).

Target I can use a scale such as ¼ in : 1 ft to find real lengths and areas from a drawing, and drawing lengths from real ones.

MTR **MTR.1.1** — Example 2 and its Your Turn: a distance, then a time · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.7.1** — every board is a plan, a map or a model: the scale is a real object's, and the answer is checked against what a room or a car could be

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A scale with two units is a rate: write it per 1 unit of drawing first.
2. Drawing × rate is real; real ÷ rate is drawing.
3. Area comes from the real dimensions — the rate is used twice, never once.

Slide by Slide

1. **Real-World Scale Drawings.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the move Notes I makes on a scale: per 1 unit first. *1 retrieves 5.02–03 — area × k^2 (yesterday). 2 retrieves 5.01 — copy ÷ original; k under 1 (earlier this unit). 3 retrieves 4.03 — decimal products; every board today has one (last unit). 4 retrieves MA.6.AR.3.2 — a ratio written as a unit rate; today a scale is read the same way (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 36 cm^2 2. k = 0.6 3. 3.6 4. 45 miles per hour
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the scale and its rewrite.** ¼ in : 1 ft says a quarter inch of paper stands for a foot of room.
6. **Notes.** A room 2 in by 2.5 in at 1 in : 4 ft is 8 ft by 10 ft, so 80 ft^2.
7. **Notes.** A 14-ft wall at 1 in : 4 ft: 14 ÷ 4 = 3.5 in on the paper.
8. **Example 1 — A room's real dimensions and area.** Ask for the scale per 1 inch before anyone multiplies.
9. **Example 1 — Worked.** 1 in : 4 ft; 10 ft by 12 ft; 120 ft^2.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 6 by 16; 96 ft^2.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A map distance, then a time.** Ask what the first answer is for — nobody stops at the distance.
13. **Example 2 — Worked.** 370 km, then 3.7 h: the scale is one rate and the speed is another.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 155 km, then 3.1 h.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. A plan is drawn at ½ in : 1 ft. A wall is 4 in long on it. How long is the real wall? → ***8 ft*** *2 — divided (4 × ½) [Notes I: per 1 inch first, then × 2]; 4 — read the scale as 1 : 1*

18–19. Board 2. A plan is drawn at 1 in : 3 ft. A room is 2.5 in long on it. How long is the real room? → ***7.5 ft*** *5.5 — added 3 [B1G-M 7.GR.1.5: the additive error]; 0.83 — divided [Notes I: drawing to real is × rate]*

20–21. Board 3. A plan is drawn at 1 in : 3 ft. A real wall is 21 ft long. How long is it on the plan? → ***7 in*** *63 — multiplied [Notes III: real to drawing is a division]; 18 — subtracted 3*

22–23. Board 4. A model car is built at 1 : 50 and is 8 cm long. How long is the real car? → ***A — 4 m*** *(B) 8 ÷ 50: divided [Notes I: drawing to real is × rate]; (C) 400 with the wrong unit: 400 cm, not 400 m [Notes I: the scale's units are the drawing's]; (D) 8 + 50: the additive error [B1G-M 7.GR.1.5] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. A plan is drawn at ¼ in : 1 ft. A real wall is 15 ft long. How long is it on the plan? → ***3.75 in*** *60 — multiplied by 4 [Notes III: real to drawing is ÷]; 7.5 — divided by 2*

26–27. Board 6. A plan is drawn at 1 in : 4 ft. A room is 2 in by 3 in on it. What is the room's real area? → ***B — 96 ft²*** *(A) 6 in² × 4: the drawing's area scaled by the length rate [B1G-M 7.GR.1.5: scaling area like length; Notes II]; (C) 2(8 + 12): the real perimeter, not the area [5.02–03 Notes I]; (D) 8 + 12: the real dimensions added [B1G-M 7.GR.1.5: the additive error] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A plane is 48 ft long. One model of it is built at 1 : 48, another at 1 : 72. Which model is longer, and by how much? → ***The 1 : 48 model, by 4 in (12 in against 8 in)*** *the 1 : 72 model — 'the bigger scale is the bigger model' [Notes III: real ÷ rate — a bigger rate divides more]; 1 ft and 0.67 ft with no comparison — stopped early*

30–31. Board 8. Mia, who is 12, enlarges her 9 in by 12 in poster into a mural at 1 in : 2 ft. What are the mural's dimensions? → ***18 ft by 24 ft*** *The 12 (her age) is not needed. 4.5 ft by 6 ft — divided [Notes I: drawing to real is × rate]; 18 in by 24 in — kept the drawing's unit*

32–33. Board 9. Andrew's drawing shows a wall 3.5 in wide; the real wall is 26.25 ft wide. The drawn height is 4 in. Find the real height and the wall's real area, and say which step uses the scale factor. → ***30 ft high; 787.5 ft². The scale factor (7.5 ft per in) is the step from 4 in to 30 ft.*** *105 ft² — 3.5 × 4 = 14 in², then × 7.5 [B1G-M 7.GR.1.5: scaling area like length]; 'the area step is the scale factor' — the scale is between a drawing and a real length, not two real ones*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 2 — divided (4 × ½) [Notes I: per 1 inch first, then × 2]

Board 2. 5.5 — added 3 [B1G-M 7.GR.1.5: the additive error]

Board 3. 63 — multiplied [Notes III: real to drawing is a division]

Board 4. 8 ÷ 50: divided [Notes I: drawing to real is × rate]

Board 5. 60 — multiplied by 4 [Notes III: real to drawing is ÷]

Board 6. 6 in² × 4: the drawing's area scaled by the length rate [B1G-M 7.GR.1.5: scaling area like length

Board 7. the 1 : 72 model — 'the bigger scale is the bigger model' [Notes III: real ÷ rate — a bigger rate divides more]

Board 8. 4.5 ft by 6 ft — divided [Notes I: drawing to real is × rate]

Board 9. 105 ft² — 3.5 × 4 = 14 in², then × 7.5 [B1G-M 7.GR.1.5: scaling area like length]

Materials Whiteboards and markers. A printed floor plan under the document camera for Notes II if the slide's rectangle is not enough. Calculators are allowed on everything and printed on nothing.

Converting Length Across Systems; Currency

Teacher's Edition

The Period

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Must Convert a length between customary and metric with a printed reference-sheet ratio, either direction, in a chain; convert money with a given rate, both ways. (MA.7.AR.3.3.)

Must not A ratio not on the FAST reference sheet; a real exchange rate the student must know; mass, volume or area (5.08–09).

Target I can convert a length between the customary and metric systems with a ratio from the reference sheet, and convert money with a given rate.

MTR **MTR.6.1** — Notes II and board 6: the estimate before the arithmetic · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote on which ratio · **MTR.3.1** — boards 5–6: two conversions in a row, each one a form of 1, with the units cancelling on the paper

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A conversion ratio is a form of 1: multiply by it and only the unit changes.
2. Pick the ratio whose 1 is the unit you have. Estimate first — which unit is bigger?
3. Money is the same move with a rate someone sets, printed on the problem.

Slide by Slide

1. **Converting Length Across Systems; Currency.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the move: a ratio, and a division because feet are bigger. *1 retrieves 5.04 — real ÷ rate (last class). 2 retrieves 5.01 — copy ÷ original; k under 1 (last week). 3 retrieves 4.02 — decimal quotients (last unit). 4 retrieves MA.6.AR.3.5 — converting inside one system; today the ratio crosses systems (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 2.5 in 2. k = ½ 3. 5 4. 2.5 ft
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the ratio and the line.** 1 in = 2.54 cm, so 2.54 cm over 1 in is 1 — the top and bottom are the same length in two units.
6. **Notes.** The sheet gives both directions: 1 in = 2.54 cm and 1 cm = 0.39 in.
7. **Notes.** An exchange rate is a conversion ratio with a date on it: 0.92 euros per dollar today, something else tomorrow, so it is printed on the problem and never on the sheet.
8. **Example 1 — Inches to centimetres.** Ask which unit should give the bigger number before anyone multiplies.
9. **Example 1 — Worked.** 38.1 cm.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 63.5 cm.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Dollars to euros, and back.** Ask what the second question's operation is before anyone writes it.
13. **Example 2 — Worked.** 73.60 euros; $50.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 162 CAD; $60.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. 1 in = 2.54 cm. A ruler is 20 in long. How many centimetres is that? → ***50.8 cm*** *7.87 — divided by 2.54 [Notes II: centimetres are smaller, so more of them]; 22.54 — added the ratio*

18–19. Board 2. 1 cm = 0.39 in. A phone is 20 cm long. How many inches is that? → ***7.8 in*** *50.8 — multiplied by 2.54 [Notes II: pick the ratio whose 1 is the unit you have]; 0.78 — place value*

20–21. Board 3. 1 mi = 1.61 km. A race is 5 mi long. How many kilometres is that? → ***8.05 km*** *3.1 — multiplied by 0.62, the other ratio [Notes II]; 6.61 — added 1.61*

22–23. Board 4. 1 km = 0.62 mi. A trail is 5 km long. How many miles is that? → ***D — 3.1 mi*** *(A) 5 + 0.62: the additive error [B1G-M]; (B) place value: 5 × 0.62 is 3.1, not 0.31 [4.03]; (C) × 1.61, the ratio for the other direction [B1G-M 7.AR.3.3: the inverted ratio; Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. 1 ft = 12 in and 1 in = 2.54 cm. A door is 6 ft tall. How many centimetres is that? → ***182.88 cm*** *15.24 — 6 × 2.54, the feet treated as inches*

[Notes I: the unit you have must match the ratio's 1]; 183 — rounded early (accept with the work shown)

26–27. Board 6. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. A table is 3 yd long. About how many centimetres is that? Estimate, then compute. → ***About 270; exactly 274.32 cm*** *7.62 — 3 × 2.54, the yards treated as inches [Notes II: estimate first — a 3-yd table is not 8 cm]; 91.44 — stopped at 36 in × 2.54 (one yard)*

28–29. Board 7. Rates on 2 Nov: $1 = 1.32 CAD; 1 CAD = 16.2 pesos. Theo changes $200 to CAD, then to pesos. How many pesos does he get? → ***4,276.80 pesos*** *3,240 — 200 × 16.2, the first hop skipped [Notes III: the rate's 1 must be the currency you have]; 2,454.55 — divided by 1.32 on the first hop*

30–31. Board 8. 1 km = 0.62 mi. Theo, who weighs 110 lb, runs a 5 km race. How long is the race in miles? → ***3.1 mi*** *The 110 lb is not needed. 8.05 — × 1.61 [B1G-M 7.AR.3.3: the inverted ratio]; 49.94 kg — converted the weight instead*

32–33. Board 9. 1 in = 2.54 cm and 1 cm = 0.39 in. Jo converts 30 cm to inches by multiplying by 2.54 and gets 76.2 in. Name the error and fix it. → ***She used the inches ratio on centimetres. Inches are bigger, so fewer: 11.7 in.*** *'She should have used 0.39' with no reason — the fix without the error named; 76.2 defended as 'more inches' — the estimate backwards [B1G-M 7.AR.3.3]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 7.87 — divided by 2.54 [Notes II: centimetres are smaller, so more of them]

Board 2. 50.8 — multiplied by 2.54 [Notes II: pick the ratio whose 1 is the unit you have]

Board 3. 3.1 — multiplied by 0.62, the other ratio [Notes II]

Board 4. 5 + 0.62: the additive error [B1G-M]

Board 5. 15.24 — 6 × 2.54, the feet treated as inches [Notes I: the unit you have must match the ratio's 1]

Board 6. 7.62 — 3 × 2.54, the yards treated as inches [Notes II: estimate first — a 3-yd table is not 8 cm]

Board 7. 3,240 — 200 × 16.2, the first hop skipped [Notes III: the rate's 1 must be the currency you have]

Board 8. 8.05 — × 1.61 [B1G-M 7.AR.3.3: the inverted ratio]

Board 9. 'She should have used 0.39' with no reason — the fix without the error named

Materials Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (they are printed on every item as well). Calculators are allowed on everything and printed on nothing.

Converting Mass, Area and Volume

Teacher's Edition

The Period

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Must Convert mass and volume with a printed reference-sheet ratio, either direction; convert area by squaring the length ratio (and volume by cubing it, in writing); route fluid ounces through cups. (MA.7.AR.3.3.)

Must not A ratio not on the FAST reference sheet (no fl oz → mL, no ft^2 → m^2 as one number); a density or a weight-from-volume question; the capacity of a solid from its dimensions (Unit 6).

Target I can convert mass, volume and area between systems, and square the length ratio for area.

MTR **MTR.5.1** — Notes II: yesterday's k^2 and today's 144 are the same structure, and the room is asked to say so · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.6.1** — every mass and volume board says its estimate first

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Mass and volume convert like length: the sheet's ratio, the estimate first.
2. Square units, square the ratio: 1 ft^2 is 144 in^2, 1 m^2 is 10,000 cm^2.
3. Fluid ounces go through cups — read the sheet for the path.

Slide by Slide

1. **Converting Mass, Area and Volume.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes II: a square foot is 12 by 12 inches, 144 of them, not 12. *1 retrieves 5.06–07 — × the ratio whose 1 is inches (yesterday). 2 retrieves 5.02–03 — area × k^2; today's 144 is the same idea (earlier this unit). 3 retrieves 4.01 — a signed decimal product (last unit). 4 retrieves MA.6.GR.2.1 — 1 ft^2 is a 12 by 12 square; today's Notes II (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 30.48 cm 2. 63 cm^2 3. −10 4. 144 in^2
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the two lines.** 1 lb = 0.454 kg is a form of 1 like 1 in = 2.54 cm.
6. **Notes.** 1 ft = 12 in, but a square foot is a 12 by 12 square: 144 in^2.
7. **Notes.** The sheet has 1 c = 8 fl oz and 1 c = 0.24 L, and no line from fluid ounces to millilitres — so a fluid ounce reaches litres through cups: a 12-fl-oz can is 12 ÷ 8 = 1.5 c, and 1.5 × 0.24 = 0.36 L.
8. **Example 1 — Pounds to kilograms.** Ask which is heavier, a pound or a kilogram, before anyone multiplies.
9. **Example 1 — Worked.** 22.7 kg.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 40.86 kg — 40.9 to one place.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Square feet to square inches.** Ask how many times the ratio is used before anyone multiplies.
13. **Example 2 — Worked.** 432 in^2.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 20,000 cm^2.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. 1 kg = 2.204 lb. A bag of rice is 10 kg. How many pounds is that? → ***22.04 lb*** *4.54 — × 0.454, the other ratio [B1G-M 7.AR.3.3: the inverted ratio]; 12.204 — added*

18–19. Board 2. 1 gal = 3.785 L. A cooler holds 4 gal. How many litres is that? → ***15.14 L*** *1.06 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 7.785 — added*

20–21. Board 3. 1 L = 0.264 gal. A tank holds 10 L. How many gallons is that? → ***2.64 gal*** *37.85 — × 3.785 [Notes I: pick the ratio whose 1 is the unit you have]; 0.264 — forgot the 10*

22–23. Board 4. 1 yd = 3 ft. A patch of lawn is 2 yd^2 in area. How many square feet is that? → ***B — 18 ft^2*** *(A) 2 × 6: 'two dimensions, so double the ratio' [5.02–03 Notes II: k × k, not 2k]; (C) 3^2 alone: the 2 yd^2 forgotten — one square yard, not two [4.06: finish the computation]; (D) 2 × 3: the ratio used once [B1G-M 7.AR.3.3: square units need the ratio squared; Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. 1 c = 8 fl oz and 1 c = 0.24 L. A bottle holds 16 fl oz. How many litres is that? → ***0.48 L*** *3.84 — 16 × 0.24, the fluid ounces treated as cups [Notes III:*

the sheet's path is fl oz → c → L]; 453.6 — used 28.35, the weight ounce [vocabulary: oz and fl oz are two units]

26–27. Board 6. 1 m = 100 cm. How many square centimetres is 1 m^2? → ***C — 10,000 cm^2*** *(A) 10 × 100: the ratio once, then a guess upward [B1G-M 7.AR.3.3: square units need the ratio squared]; (B) 2 × 100: 'two dimensions, so double it' [5.02–03 Notes II]; (D) the length ratio handed in for an area [B1G-M 7.AR.3.3: square units need the ratio squared; Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. 1 kg = 2.204 lb. A 5-lb bag of flour and a 2-kg bag of sugar. Which is heavier, and by how much? → ***The 5-lb bag, by 0.592 lb (about 0.6 lb)*** *'the 5-lb bag, because 5 > 2' — the numbers compared, not the weights [MTR.6.1: units first]; 'the 2-kg bag' — kilograms taken as heavier than any number of pounds*

30–31. Board 8. 1 gal = 3.785 L. A 20-gal fish tank, bought for $45, is being filled. How many litres does it hold? → ***75.7 L*** *The $45 is not needed. 5.28 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 170.33 — the price used*

32–33. Board 9. 1 ft = 12 in. Priya says 2 ft^3 is 24 in^3, because 2 × 12 = 24. Name the error and fix it. → ***She used the ratio once. A cubic foot is 12 × 12 × 12 = 1,728 in^3, so 2 ft^3 = 3,456 in^3.*** *288 in^3 — the ratio squared, today's rule applied to a cube [Notes III: three lengths, three times]; 'it should be 24 ft' — the unit changed instead of the number*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 4.54 — × 0.454, the other ratio [B1G-M 7.AR.3.3: the inverted ratio]

Board 2. 1.06 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]

Board 3. 37.85 — × 3.785 [Notes I: pick the ratio whose 1 is the unit you have]

Board 4. 2 × 6: 'two dimensions, so double the ratio' [5.02–03 Notes II: k × k, not 2k]

Board 5. 3.84 — 16 × 0.24, the fluid ounces treated as cups [Notes III: the sheet's path is fl oz → c → L]

Board 6. 10 × 100: the ratio once, then a guess upward [B1G-M 7.AR.3.3: square units need the ratio squared]

Board 7. 'the 5-lb bag, because 5 > 2' — the numbers compared, not the weights [MTR.6.1: units first]

Board 8. 5.28 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]

Board 9. 288 in^3 — the ratio squared, today's rule applied to a cube [Notes III: three lengths, three times]

Materials Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (printed on every item as well); a 12 by 12 grid under the document camera for Notes II. The Unit Review paper to send home. Calculators are allowed on everything and printed on nothing.

Multi-Step Proportion Problems

Teacher's Edition

The Period

MA.7.AR.4.5 Solve real-world problems involving proportional relationships.

Must Solve a two- or three-step proportional problem (a rate used twice; a recipe scaled and checked; a part-to-whole mixture), and compare two rates after putting them in one unit with a printed reference-sheet ratio. (MA.7.AR.4.5, with MA.7.AR.3.3 inside it.)

Must not A graph, a table of a proportional relationship, or y = px named as such (MA.7.AR.4.1–4.4, Unit 7); a ratio not on the FAST sheet; a percent problem (Unit 8).

Target I can solve a problem that takes two or more proportional steps, including a unit conversion, and compare two rates given in different units.

MTR **MTR.1.1** — Notes I and board 4: a problem whose first answer (miles per tank) is not the answer, and the room keeps going · **MTR.6.1** — Notes II and board 6: 'which is faster?' is refused until the units match; the answer is checked against the estimate that a mile is longer · **MTR.7.1** — boards 8–9: a plan's area in square metres and a choice of car

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A multi-step problem is one unit rate used again — name what each step finds.
2. Same unit first; then compare.
3. A recipe's k comes from the yields; then every ingredient, then the pantry.

Slide by Slide

1. **Multi-Step Proportion Problems.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the unit rate every board today starts from. *1 retrieves 5.08–09 — × the ratio whose 1 is kilograms (yesterday). 2 retrieves 5.04 — drawing × rate (earlier this unit). 3 retrieves 4.04 — a two-step with decimals (last unit). 4 retrieves MA.6.AR.3.2 — a unit rate; today every board starts from one (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 8.816 lb 2. 150 km 3. 9 4. 60 miles per hour
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the two lines.** A car does 23 miles per gallon and its tank holds 17.5 gal: 23 × 17.5 = 402.5 mi on a tank.
6. **Notes.** 100 km/h against 65 mi/h: nothing compares until the units match.
7. **Notes.** 12 cookies take 1½ cups of flour.
8. **Example 1 — Two rates in different units.** Ask which unit the comparison will be made in before anyone computes.
9. **Example 1 — Worked.** A: 8.05 km in ⅔ h is 12.075 km/h; B: 9 km in ¾ h is 12 km/h.
10. **Your Turn — Same steps, your numbers. Boards up when done.** A: 6.44 km in ½ h is 12.88 km/h; B: 7 km in 7⁄12 h is 12 km/h.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A recipe scaled, then checked.** Ask what k is before anyone touches the rice.
13. **Example 2 — Worked.** k = 2.5, so 6¼ cups; 5 cups is 2 recipes, 16 servings.
14. **Your Turn — Same steps, your numbers. Boards up when done.** k = 2, 3 cups; 4 cups is 2⅔ recipes, 16 servings.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** A car does 28 miles per gallon. Its tank holds 15 gal. How far does it go on a full tank? → ***420 mi*** *1.87 — 28 ÷ 15 [Notes I: miles per gallon × gallons gives miles]; 43 — added*
17. **18–19. Board 2.** A printer prints 35 pages per minute. A job is 1,400 pages. How long does the job take? → ***40 min*** *49,000 — multiplied [Notes I: name what the step finds — pages × pages/min is not minutes]; 400 — place value*
18. **20–21. Board 3.** Mia earns $14.50 per hour at a shop 2 mi from home, 6 h a day for 5 days. How much does she earn? → ***$435*** *The 2 mi is not needed. 87 — one day's pay [Notes I: the question asks for five days]; 2,175 — 435 × 5, the days counted twice*
19. **22–23. Board 4.** A car does 30 miles per gallon with a 12-gal tank. It starts a 900-mi trip with a full tank. How many tanks of gas does the trip take? → ***D — 3 tanks*** *(A) 2.5 rounded down: half a tank of road left undriven [Notes I: a part of a tank is still a tank to buy]; (B) 900 ÷ 12: the gallons used as miles per tank [Notes I: the rate first]; (C) 900 ÷ 36: the two numbers added before dividing [B1G-M 7.AR.4.5: steps not named] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. A cereal has 12 g of protein per 100 g. How many grams of protein in 1 kg? → ***120 g*** *12 — the rate copied with the wrong unit [Notes I: per 100 g, and a kilogram is ten of those]; 1,200 — × 100*

26–27. Board 6. 1 mi = 1.61 km. Cyclist A rides 12 km in 50 min; cyclist B rides 7 mi in 45 min. Who is faster? → ***C — cyclist B*** *(A) 12 km and 12 km/h confused: a distance read as a speed [B1G-M 7.AR.4.5: a rate is per unit of time]; (B) 12 > 7, the distances compared in two units [Notes II: same unit first]; (D) 11.27 is B's distance in km, not a speed — the time was never used [B1G-M 7.AR.4.5: a rate is per unit of time] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. Paint is mixed white : blue : grey = 2 : 4 : 5. A batch uses 8 gal of white. How many gallons is the whole batch? → ***44 gal*** *20 — 8 + 4 + 5 ×1, the ratio read as gallons [Notes III: k from the part you know]; 32 — the grey (5 × 4) and nothing more*

30–31. Board 8. A plan is drawn at 1 in : 4 ft; a room is 2 in by 2.5 in on it. 1 ft = 0.305 m. Find the room's area in square metres. → ***About 7.44 m²*** *24.4 — 80 ft² × 0.305, the ratio once on an area [5.08–09 Notes II]; 20 — 2 × 2.5 × 4, the drawing's area scaled once [5.04 board 6]*

32–33. Board 9. 1 mi = 1.61 km. A 300-km drive. Car A averages 50 mi/h; car B averages 75 km/h. Which car gets there first, and by how long? Explain. → ***Car A: 80.5 km/h against 75, so about 3.73 h against 4 h — about 16 minutes sooner.*** *'Car B, 75 > 50' — numbers compared in two units [Notes II: same unit first]; 'Car A, 80.5 > 75' with no times — the decision without the 'by how long'*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 1.87 — 28 ÷ 15 [Notes I: miles per gallon × gallons gives miles]

Board 2. 49,000 — multiplied [Notes I: name what the step finds — pages × pages/min is not minutes]

Board 3. 87 — one day's pay [Notes I: the question asks for five days]

Board 4. 2.5 rounded down: half a tank of road left undriven [Notes I: a part of a tank is still a tank to buy]

Board 5. 12 — the rate copied with the wrong unit [Notes I: per 100 g, and a kilogram is ten of those]

Board 6. 12 km and 12 km/h confused: a distance read as a speed [B1G-M 7.AR.4.5: a rate is per unit of time]

Board 7. 20 — 8 + 4 + 5 ×1, the ratio read as gallons [Notes III: k from the part you know]

Board 8. 24.4 — 80 ft² × 0.305, the ratio once on an area [5.08–09 Notes II]

Board 9. 'Car B, 75 > 50' — numbers compared in two units [Notes II: same unit first]

Materials Whiteboards and markers. The FAST reference sheet's conversion lines on the board (printed on every item as well). The Unit Review key to post. Calculators are allowed on everything and printed on nothing. 43 minutes: the timings below are the short ones.